

Pon Aung

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Portfolio: PonAppliedPhysics.github.io

EDUCATION

The Ohio State University

Columbus, OH

Bachelor of Science in Physics

Minor: Engineering Science, Mechanical Engineering Emphasis

Coursework: Statics, Mechanics of Materials, Dynamics, Material Science, Intermediate Electricity and Magnetism, Experimental Physics Instrumentation, Optics, AI for Mechanical Engineer (University of Michigan)

EXPERIENCE

Honda Development and Manufacturing of America

Marysville, OH

Battery Safety Systems Project Lead – BEV Thermal Runaway Detection Capstone

August 2024 – May 2025

- Evaluated 9 thermal runaway detection technologies using weighted decision analysis to support mechanical design and thermal systems safety requirements for Honda BEV battery packs
- Designed ultrasonic pulse-echo experiments using 1 MHz transducers to characterize battery charge states; ANOVA confirmed statistically significant voltage variation during charging and discharging cycles
- Evaluated infrared pyrometer accuracy against FLIR thermal measurements at multiple distances, identifying surface emissivity effects that reduced detection reliability on reflective aluminum
- Validated HF gas sensor detection at 0.5, 1.5, and 2.5 m with coefficient of variation under 4 percent for timing and under 8 percent for concentration, confirming sub-1 ppm sensitivity at the longest test distance
- Delivered sensor validation data, selection matrix, and layered multi-sensor deployment strategy to Honda engineering leadership, completing all testing at 36 percent of the \$3,500 budget

PROJECT EXPERIENCE

OSU Underwater Robotics – Talos AUV

Columbus, OH

Mechanical Team Member

December 2025 – May 2026

- Developed CAD models and mechanical designs in SolidWorks for hull and chassis components; supported rapid prototyping and manufacturing processes using mills, lathes, and 3D printing to integrate a dual stereo-camera vision system into the Talos AUV for RoboSub
- Redesigned Delrin torpedo launcher assemblies through iterative design, tolerance analysis, and DFM principles, integrating mechanical seals and custom gearbox mechanisms that eliminated water-ingress failures from the prior RoboSub design
- Led electromechanical integration and subsystem integration efforts through iterative pool testing, adding structural reinforcement and guided alignment features that improved torpedo launch accuracy and mechanical reliability over the prior design

Computational Heat Transfer Analysis and Experimental Validation

Columbus, OH

Physics 5700 – Advanced Experimental Physics [Github](#)

Spring 2026

- Validated 11-thermocouple sensor array data through systematic calibration and uncertainty analysis, measuring and correcting per-sensor offsets ($\sim 0.6^{\circ}\text{C}$) to verify cross-run consistency across independent thermal system experiments
- Developed and optimized 5 finite-difference thermal models using data-driven boundary conditions and parameter optimization, reducing chi-squared error by $10\times$ while decreasing runtime from minutes to 3 seconds through vectorized NumPy implementation
- Built a physics-informed neural network in TensorFlow to model heat transfer behavior and recover thermal diffusivity directly from experimental sensor data, achieving MAE of 0.42°C without manual parameter tuning

TECHNICAL SKILLS

CAD & Mechanical Design: SolidWorks, Shapr3D, CAD modeling, mechanical design, prototyping, tolerance analysis, DFM, technical drawings

Manufacturing & Fabrication: 3D printing, machining, mills, lathes, manufacturing processes, subsystem assembly, lab instrumentation

Programming & Simulation: MATLAB, Python (NumPy, SciPy, Pandas, Matplotlib, TensorFlow)

Analysis & Modeling: Finite-difference PDE solvers, parameter optimization, heat transfer modeling, stability analysis, uncertainty analysis, weighted least-squares regression, ANOVA, chi-squared goodness-of-fit testing

Mechanical Engineering: Heat transfer, statics, dynamics, structural analysis, circuits, sensor validation, lab instrumentation and data collection